

Assignment No.1

1. Create 'spj' database'. (Database for suppliers, parts and jobs) 2. use spj database; 3. Create following tables in spj database.

```
Command Prompt - mysql -u X
owners.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
kd2_98837_sakshi> create database spj;
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> CREATE TABLE S('S#' CHAR(5) NOT NULL,Sname CHAR(20),status SMALLINT,city CHAR(15));
ERROR 1046 (3D000): No database selected
kd2_98837_sakshi> SELECT SPJ;
ERROR 1054 (42S22): Unknown column 'SPJ' in 'field list'
kd2_98837_sakshi> SELECT spj;
ERROR 1054 (42S22): Unknown column 'spj' in 'field list'
kd2_98837_sakshi> USE SPJ
Database changed
kd2_98837_sakshi> CREATE TABLE S('S#' CHAR(5) NOT NULL,Sname CHAR(20),status SMALLINT,city CHAR(15));
Query OK, 0 rows affected (0.03 sec)

kd2_98837_sakshi> INSERT INTO S VALUES('S1','SMITH',20,'LONDON');
ERROR 1054 (42S22): Unknown column 'S1' in 'field list'
kd2_98837_sakshi> INSERT INTO S VALUES('S1','SMITH',20,'LONDON');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO S VALUES('S2','JONES',10,'PARIS');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO S VALUES('S3','BLAKE',30,'PARIS');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO S VALUES('S4','CLARK',20,'LONDON');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO S VALUES('S5','ADAMS',30,'ATHENS');
Query OK, 1 row affected (0.01 sec)
```

```
Command Prompt - mysql -u < + >
kd2_98837_sakshi> CREATE TABLE P ('P#' CHAR(6) NOT NULL, Pname CHAR(20), color CHAR(6), weight SMALLINT, city CHAR(15));
Query OK, 0 rows affected (0.03 sec)

kd2_98837_sakshi> INSERT INTO P VALUES('P1','NUT','RED',12,'LONDON');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO P VALUES('P2','BOLT','GREEN',17,'PARIS');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO P VALUES('P3','SCREW','BLUE',17,'ROME');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO P VALUES('P4','SCREW','RED',14,'LONDON');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO P VALUES('P5','CAM','BLUE',12,'PARIS');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO P VALUES('P6','COG','RED',19,'LONDON');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> SELECT * FROM P;
+-----+-----+-----+-----+-----+
| P# | Pname | color | weight | city |
+-----+-----+-----+-----+-----+
| P1 | NUT   | RED   | 12     | LONDON |
| P2 | BOLT  | GREEN | 17     | PARIS  |
| P3 | SCREW | BLUE  | 17     | ROME   |
| P4 | SCREW | RED   | 14     | LONDON |
| P5 | CAM   | BLUE  | 12     | PARIS  |
| P6 | COG   | RED   | 19     | LONDON |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

kd2_98837_sakshi> SELECT * FROM S;

+-----+-----+-----+-----+
| S# | Sname | status | city |
+-----+-----+-----+-----+
| S1 | SMITH | 20     | LONDON |
| S2 | JONES | 10     | PARIS  |
| S3 | BLAKE | 30     | PARIS  |
| S4 | CLARK | 20     | LONDON |
| S5 | ADAMS | 30     | ATHENS |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

kd2_98837_sakshi> CREATE TABLE J ('J#' CHAR(4) NOT NULL, Jname CHAR(10), City CHAR(15));
Query OK, 0 rows affected (0.03 sec)

kd2_98837_sakshi> INSERT INTO J VALUES('J1','SORTER','PARIS');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO J VALUES('J2','PUNCH','ROME');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO J VALUES('J3','READER','ATHENS');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO J VALUES('J4','CONSOLE','ATHENS');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO J VALUES('J5','COLLATOR','LONDON');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO J VALUES('J6','TERMINAL','OSLO');
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO J VALUES('J7','TAPE','LONDON');
```

```
Command Prompt - mysql -u x + v
kd2_98837_sakshi> SELECT * FROM J;
+----+-----+-----+
| J# | Jname | City |
+----+-----+-----+
| J1 | SORTER | PARIS |
| J2 | PUNCH  | ROME  |
| J3 | READER  | ATHENS |
| J4 | CONSOLE | ATHENS |
| J5 | COLLATOR | LONDON |
| J6 | TERMINAL | OSLO |
| J7 | TAPE    | LONDON |
+----+-----+-----+
7 rows in set (0.00 sec)

kd2_98837_sakshi> CREATE TABLE sp ('S#' CHAR(4) NOT NULL, 'P#' CHAR(4) NOT NULL, 'J#' CHAR(4) NOT NULL, QTY INT);
Query OK, 0 rows affected (0.03 sec)

kd2_98837_sakshi> INSERT INTO P VALUES('S1','P1','J1',200);
ERROR 1136 (21S01): Column count doesn't match value count at row 1
kd2_98837_sakshi> INSERT INTO SP VALUES('S1','P1','J1',200);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S1','P1','J4',700);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S2','P3','J1',400);
Query OK, 1 row affected (0.00 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S2','P3','J2',200);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S2','P3','J3',200);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S2','P3','J4',500);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S2','P3','J5',600);
Query OK, 1 row affected (0.00 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S2','P3','J6',400);
Query OK, 1 row affected (0.00 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S2','P3','J7',800);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S2','P5','J2',100);
Query OK, 1 row affected (0.00 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S3','P3','J1',200);
Query OK, 1 row affected (0.00 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S3','P4','J2',500);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S4','P6','J3',300);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S4','P6','J7',300);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S5','P2','J2',200);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S5','P2','J4',100);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S5','P5','J5',500);
```

```
Command Prompt - mysql -u x + v
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S5','P5','J7',100);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S5','P6','J2',200);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S5','P1','J4',100);
Query OK, 1 row affected (0.00 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S5','P3','J4',200);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S5','P4','J4',800);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S5','P5','J4',400);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> INSERT INTO SP VALUES('S5','P6','J4',500);
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> SELECT * FROM SP;
+-----+
| S# | P# | J# | QTY |
+-----+
| S1 | P1 | J1 | 200 |
| S1 | P1 | J4 | 700 |
| S2 | P3 | J1 | 400 |
| S2 | P3 | J2 | 200 |
| S2 | P3 | J3 | 200 |
| S2 | P3 | J4 | 500 |
| S2 | P3 | J5 | 600 |
| S2 | P3 | J6 | 400 |
+-----+
```

```
Command Prompt - mysql -u x + v
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> SELECT * FROM SP;
+-----+
| S# | P# | J# | QTY |
+-----+
| S1 | P1 | J1 | 200 |
| S1 | P1 | J4 | 700 |
| S2 | P3 | J1 | 400 |
| S2 | P3 | J2 | 200 |
| S2 | P3 | J3 | 200 |
| S2 | P3 | J4 | 500 |
| S2 | P3 | J5 | 600 |
| S2 | P3 | J6 | 400 |
| S2 | P3 | J7 | 800 |
| S2 | P5 | J2 | 100 |
| S3 | P3 | J1 | 200 |
| S3 | P4 | J2 | 500 |
| S4 | P6 | J3 | 300 |
| S4 | P6 | J7 | 300 |
| S5 | P2 | J2 | 200 |
| S5 | P2 | J4 | 100 |
| S5 | P5 | J5 | 500 |
| S5 | P5 | J7 | 100 |
| S5 | P6 | J2 | 200 |
| S5 | P1 | J4 | 100 |
| S5 | P3 | J4 | 200 |
| S5 | P4 | J4 | 800 |
| S5 | P5 | J4 | 400 |
| S5 | P6 | J4 | 500 |
+-----+
24 rows in set (0.00 sec)

kd2_98837_sakshi> |
```

Q2.

1. Create 'hr' database. Import hr-db.sql.
2. Create 'sales' database. Import sales-db.sql.
3. List all tables in hr database. Display contents of all tables (one by one).
4. List all tables in sales database. Display contents of all tables (one by one).

```
Command Prompt - mysql -u X Command Prompt - mysql -u X Command Prompt - mysql -u X + v X
root> create database hr;
Query OK, 1 row affected (0.01 sec)

root> use hr;
Database changed
root> source D:/Sunbeam_git_data/DBT/dbs/new-hr-db.sql
Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
```

1.

```
Command Prompt - mysql -u X Command Prompt - mysql -u X Command Prompt - mysql -u X + v X
Query OK, 0 rows affected (0.01 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)
Query OK, 0 rows affected (0.00 sec)

root> show tables;
+-----+
| Tables_in_hr |
+-----+
| countries    |
| departments  |
| employees    |
| job_history  |
| jobs         |
| locations    |
| regions      |
+-----+
7 rows in set (0.00 sec)
```

```
Command Prompt - mysql -u x Command Prompt - mysql -u x Command Prompt - mysql -u x + v
root> create database joins;
Query OK, 1 row affected (0.01 sec)

root> use joins;
Database changed
root> source D:/Sunbeam_git_data/DBT/dbs/joins.sql
Query OK, 0 rows affected, 1 warning (0.01 sec)

Query OK, 0 rows affected, 1 warning (0.00 sec)
Query OK, 0 rows affected, 1 warning (0.00 sec)
Query OK, 0 rows affected, 1 warning (0.00 sec)
Query OK, 0 rows affected, 1 warning (0.00 sec)
Query OK, 0 rows affected (0.03 sec)
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.00 sec)
Query OK, 1 row affected (0.00 sec)
Query OK, 1 row affected (0.00 sec)
Query OK, 0 rows affected (0.04 sec)
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.00 sec)
Query OK, 1 row affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)

root> show tables;
+-----+
| Tables_in_joins |
+-----+
| addr             |
| depts            |
| emp_meeting     |
| emps            |
| meeting          |
+-----+
5 rows in set (0.00 sec)

root> create database classwork;
ERROR 1007 (HY000): Can't create database 'classwork'; database exists
root> create database classworks;
Query OK, 1 row affected (0.01 sec)

root> use classworks;
Database changed
root> source D:/Sunbeam_git_data/DBT/dbs/classwork-db.sql
Query OK, 0 rows affected, 1 warning (0.00 sec)

Query OK, 0 rows affected, 1 warning (0.00 sec)
Query OK, 0 rows affected, 1 warning (0.00 sec)
Query OK, 0 rows affected, 1 warning (0.00 sec)
Query OK, 0 rows affected, 2 warnings (0.04 sec)
Query OK, 1 row affected (0.01 sec)
```

```
root> show tables;
+-----+
| Tables_in_classworks |
+-----+
| books                |
| dept                 |
| emp                  |
| salgrade              |
+-----+
4 rows in set (0.00 sec)

root>
```

[illegible]

```
Command Prompt - mysql -u X Command Prompt - mysql -u X Command Prompt - mysql -u X + v
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.00 sec)
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.01 sec)
Query OK, 1 row affected (0.01 sec)

root> show tables;
+-----+
| Tables_in_sales |
+-----+
| customers       |
| orders          |
| salespeople     |
+-----+
3 rows in set (0.00 sec)

root>
```

```
Command Prompt - mysql -u X Command Prompt - mysql -u X Command Prompt - mysql -u X + v
3 rows in set (0.00 sec)

root> select * from customers;
+-----+
| cnum | cname   | city   | rating | snum |
+-----+
| 2001 | Hoffman | London | 100    | 1001 |
| 2002 | Giovanni | Rome   | 200    | 1003 |
| 2003 | Liu     | San Jose | 200    | 1002 |
| 2004 | Grass   | Berlin  | 300    | 1002 |
| 2006 | Clemens | London  | 100    | 1001 |
| 2008 | Cisneros | San Jose | 300    | 1007 |
| 2007 | Pereira | Rome    | 100    | 1004 |
+-----+
7 rows in set (0.00 sec)

root> select * from orders;
+-----+
| onum | amt   | odate   | cnum | snum |
+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 | 75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+
10 rows in set (0.00 sec)

root> select * from salespeople;
```

```
Command Prompt - mysql -u X Command Prompt - mysql -u X Command Prompt - mysql -u X + v
| 2008 | Cisneros | San Jose | 300 | 1007 |
| 2007 | Pereira | Rome | 100 | 1004 |
+-----+
7 rows in set (0.00 sec)

root> select * from orders;
+-----+
| onum | amt | odate | cnum | snum |
+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 | 75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+
10 rows in set (0.00 sec)

root> select * from salespeople;
+-----+
| snum | sname | city | comm |
+-----+
| 1001 | Peel | London | 0.12 |
| 1002 | Serres | San Jose | 0.13 |
| 1004 | Motika | London | 0.11 |
| 1007 | Rifkin | Barcelona | 0.15 |
| 1003 | Axelrod | New York | 0.10 |
+-----+
5 rows in set (0.00 sec)

root>
```



Q3.

Answer the following questions (by observations only – not queries).

1. How many orders data is present in the database?

Ans. 10 Orders are present in the database.

2. How many customers are present in the database?

Ans. 7 customers are present in the database.

3. How many salespeople are present in the database?

Ans. 5 salespeople are present in the database.

4. On which date order with highest amount is placed?

Ans. 04-10-1990 is the date order with highest amount is placed.

5. For which order salesman got maximum commission? Hint: you need to refer data of two tables.

Ans. 3011 order salesman got maximum commission.

6. Which salesmen are handling more than two customers? Hint: you need to refer data of two tables.

Ans. Peel salesmen are handling more than two customers.

7. Which city have single customer?

Ans. Berlin city have single customer.

8. Which city has multiple salespeople?

Ans. London city has multiple salespeople.

9. What is name of customer and salesman of the maximum amount order?

Ans. Clemens is name of customer and salesman of the maximum amount order.

10. Study relationship between the tables in sales database.

Ans. A) 1 salespeople have many customers.

B) 1 customer have many orders.

C) 1 salespeople have many orders.